

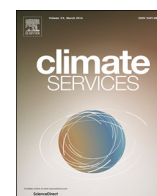


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Original research article

Global crop yield forecasting using seasonal climate information from a multi-model ensemble

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ABSTRACT

Forecasting year-to-year variations in the yields of major crops globally is expected to have utility in strengthening the ability of societies to better respond to food production shocks and food price spikes triggered by climate extremes. However, substantial improvements to the methodology used in global crop forecasting are required to realize a reliable operational service. Here, we assess the reliability of global within-season and pre-season predictions of yield variability obtained by applying statistical yield models to seasonal temperature and precipitation hindcast data derived from a multi-model ensemble (MME). This analysis is performed for five individual atmosphere-ocean coupled general circulation models (GCMs) and the two MME datasets generated using the average method and the mosaic method. Four major crops, maize, rice, wheat and soybean are studied. The mosaic method reliably predicts the yield variability over a large portion (25–38%) of the global harvested area three months before harvesting. The areas where the mosaic method displays good prediction skill are two to three times larger than those achieved using the average method (9–15%). Using the mosaic method, reliable within-season predictions of national yield variability can be produced in 36%, 24%, 25% and 30% of the maize-, soybean, rice- and wheat-producing countries, respectively. The pre-season predictions are found to be reliable in 23–32% of the crop-producing countries. As yield variability at the national level is of interest to commodity and food security specialists, the mosaic method provides a basis for developing an operational global yield forecasting service.

Practical implications

This research proposes a method that can be used to improve the reliability of global yield forecasting of major crops, ultimately allowing climate centers and research institutes to develop an operational global crop forecasting service. After the food crises of the 2000s and 2010s, tools for global agricultural monitoring based on governmental statistics and satellite remote sensing have been developed to aid in responding efficiently to such shocks. Global crop forecasting services, including the within-season and pre-season predictions of yield variability elaborated in this article, can provide independent and complementary information for use by national governments, commercial entities and international organizations because the main purpose of existing tools is monitoring, rather than forecasting. Such crop forecasting services increase the ability of societies to address the risks to food supplies posed by climate variability and change.

1. Introduction

Climate extremes, which reduce crop production, and the associated export bans can trigger food price spikes (Headey, 2011; Porter et al., 2014; Lesk et al., 2016). An example is the wheat embargo issued by the Russian government during 2010/2011 due to a drought and heat wave (Welton, 2011). As consumers in many countries are more dependent on food imports than previously (FAO, 2011), national governments and commercial entities in import-dependent countries pay close attention to variations in food production and export prices in major food-exporting countries, as well as their own domestic production (Iizumi et al., 2013). The demand for monitoring and forecasting of climate shocks, as well as providing early warnings of such shocks, on agricultural production is therefore increasing worldwide.

Crop yield forecasting based on seasonal climate forecasts provides a means of limiting the negative effects of climate variability and extremes under present-day climate conditions. At the same time, yield forecasting represents an adaptation technology to climate change. The

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Table 1

A summary of the GCMs used in this study.

Model acronym	Developer (country)	Ensemble members	References
APCC	APEC Climate Center (Korea)	10	Jeong et al. (2008)
MSC-CANCM3	Meteorological Service of Canada (Canada)	10	Scinocca et al. (2008)
NCEP	National Centers for Environmental Prediction (United States)	20	Saha et al. (2014)
NASA	National Aeronautics and Space Administration (United States)	11	Molod et al. (2012)
PNU	Pusan National University (Korea)	5	Park and Ahn (2004)

magnitudes of the year-to-year variations in seasonal average temperature and precipitation in many locations around the world are larger than their projected changes around the middle of this century (Deser et al., 2012). Experiences in responding to variations in growing season climates and the associated food production shocks undoubtedly improve the ability of societies to address the risks posed by climate change.

Some yield forecasting services cover specific regions, such as Europe, Africa, the United States (US) and China (FAO, 2016a). However, global yield forecasting has an advantage compared to regional forecasting in that it provides information on possible yield variations in both food-exporting and food-importing countries, which have implications for food agencies that attempt to maintain national food balances. Global yield forecasting based on seasonal climate forecast data is under-studied, with a few exceptions (Iizumi et al., 2013, 2016). An earlier global analysis that employed nine-member ensemble mean hindcast data of temperature and soil moisture simulated using a single GCM (Iizumi et al., 2013) concluded that yield losses of rice and wheat over 15–19% of the global harvested area can be predicted reliably three months before harvesting; however, reliable global within-season predictions for maize and soybean are not feasible because the modeled yields of these crops are more dependent on moisture conditions than temperature, and the predictability of moisture variables is in general low (Iizumi et al., 2016). While novel, this earlier study (Iizumi et al., 2013) has two main limitations, specifically the low predictability of maize and soybean yields and the limited extent of the area that displays good prediction skill (about one-fifth of the global harvested area). These limitations are crucial, as these crops are international agricultural commodities, and the limited area where skillful predictions can be produced is insufficient for food agencies in many countries. Therefore, improvements to the methodology used in global yield forecasting are required to overcome these limitations.

The multi-model ensemble (MME) technique is widely used in the climate science community to improve the reliability of weather and climate forecasts at various time scales (Krishnamurti et al., 2000, 2016; Knutti et al., 2010). This technique computes the statistics of a climatic variable of interest using ensemble forecast data from multiple GCMs. The use of the MME technique has the potential to improve global yield forecasting because earlier work (Iizumi et al., 2013, 2016) is based on the outputs of a single GCM (i.e., the Scale Interaction Experiment–Frontier Research Center for Global Change coupled GCM (SINTEX-F1); Luo et al., 2005). To test this hypothesis, the Asia-Pacific Economic Cooperation (APEC) Climate Center (APCC) offers a promising opportunity. The APCC has access to seasonal climate forecast data provided by 13 operational climate centers and research institutes in 10 of the member economies that participate in APEC and produces MME seasonal temperature and precipitation forecasts for the member economies (Min et al., 2014).

Using the outputs of five GCMs that are employed in the MME seasonal climate forecasts of the APCC, we present a global assessment of the prediction skills for yield variability at three and six months before harvesting through applying statistical yield models to the seasonal temperature and precipitation hindcast data obtained from the MME. The predictions made at lead times of three and six months are referred to as the within-season and pre-season predictions, respectively, following Iizumi et al. (2013). In this study, the term *yield*

variability indicates the sign of the yield change between a year of interest and the previous year. The magnitude of the yield change is not assessed in this study. We examine two different methods when providing MME temperature and precipitation hindcast data. The average method computes a simple average over multiple GCMs with even weights. The mosaic method selects a single best-performing GCM for each location and cropping season, based on the skill scores for yield variability. Five GCMs (APCC, MSC-CANCM3, NCEP, NASA and PNU; see Table 1 for these abbreviations) are selected for use in this study. The availability of relatively long time series (1983–2015) underlies this selection. The analysis is completed for four major crops, maize, rice, wheat and soybean, which together account for two-thirds of food calories worldwide.

2. Material and methods

2.1. Statistical yield models

The yield models used here are similar to those presented in Iizumi et al. (2013), with two differences. First, the moisture variable used here is precipitation, whereas the root-zone soil moisture was used in Iizumi et al. (2013). Second, the periods studied here (1984–2010 at the grid level and 1984–2015 at the national level) are longer than that studied in Iizumi et al. (2013), who investigated the period 1983–2006.

In the yield models, the response variable is the yield anomaly (the departure from the normal yield), and the explanatory variables are the anomalies of growing season average temperature and precipitation. Specifically, the response variable is the normalized first-difference annual yield time series of a crop:

$$\Delta Y_{t,g,s} = \frac{(Y_{t,g,s} - Y_{t-1,g,s})}{Y_{t-3:t-1,g,s}} \times 100, \quad (1)$$

where the subscript t indicates the harvesting year; g indicates the grid cell; and s indicates the cropping season of the crop. The grid-cell yield data used here were obtained from the updated version of the global dataset of historical yields (Iizumi et al., 2014b; Iizumi and Ramankutty, 2016). In this yield dataset, maize and rice have two cropping season categories, *major* and *secondary*. Wheat also has two categories, but they are referred to as *winter* and *spring*. Soybean has only one single *major* cropping season. ΔY is the yield change between year t and the previous year ($t - 1$), expressed as a percentage relative to the 3-year running mean for the period from $t - 3$ to $t - 1$, which represents normal yield. Hereafter, ΔY is referred to as the yield anomaly. Calculation of the first-difference yield time series emphasizes the yield change that is due mainly to seasonal climate, although other factors affect year-to-year variations in yield.

The first-difference annual time series are computed for the growing season average 2-m air temperature and precipitation:

$$\Delta T_{t,g,s} = T_{t,g,s} - T_{t-1,g,s}, \quad (2)$$

$$\Delta P_{t,g,s} = P_{t,g,s} - P_{t-1,g,s}, \quad (3)$$

where T and P indicate the average temperature (°C) and precipitation (mm d⁻¹) over the reproductive growth period (RGP) of a crop. Although the vegetative growth period is important, the climate during the reproductive growth period, which covers the yield formation stage,

has greater relevance in determining yields. We assume that a 3-month interval just before harvesting represents the reproductive growth period, although it may be longer than the actual period for some crops and locations. Global crop calendar data for 2000 (Sacks et al., 2010) are used to determine this interval.

A multiple linear regression model is then constructed:

$$\Delta Y_{t,g,s} = a_{0,g,s} + a_{1,g,s} \cdot \Delta T_{t,g,s} + a_{2,g,s} \cdot \Delta P_{t,g,s} + \varepsilon, \quad (4)$$

where a_0 , a_1 and a_2 indicate the regression coefficients, and ε indicates the error term, which follows a normal distribution with a mean of zero and a standard deviation of $\sigma_{\varepsilon,g,s}$. These coefficients are estimated for each crop, cropping season and grid cell using the Markov Chain Monte Carlo (MCMC) method, following Iizumi et al. (2013). Given a grid size of 1.125°, it is reasonable to estimate the coefficient values independently for each grid cell because agronomic technology, crop cultivar and management practices could differ across neighboring grid cells. Although the ordinary least squares method would work well for the multiple linear regression models, we use the MCMC method to replicate the analysis presented in Iizumi et al. (2013) as closely as possible.

For the crops except soybean, we compute the average grid-cell yield anomalies over the two cropping seasons of each crop (*major/secondary* or *winter/spring*) using the average production in the 1990s during the different seasons as the weights, following Iizumi et al. (2013). When calculating national average yield anomalies, we average the predicted grid-cell yield anomalies over each country, using the grid-cell harvested area in 2000 obtained from Monfreda et al. (2008) as the weights. Although the harvested area changes with time, historical crop-specific maps are not available.

In addition to the yield model based on temperature and precipitation (the T-P model), we use the temperature-soil moisture (T-S) model to enable comparison with the data presented in Iizumi et al. (2013). The temperature- and precipitation-only models (the T model and the P model, respectively) are also used to infer which climatic variable is more important in explaining the observed yield variability. The P model is used to examine the sensitivity of the prediction skill to irrigation.

2.2. Climate data

We use the Japanese 25-year Reanalysis (JRA25; Onogi et al., 2007) to represent actual climate conditions. The RGP average temperature and precipitation are calculated using the daily reanalysis data. We first determine the last month of the reproductive growth period:

$$m_{\text{end}} = \begin{cases} 12 & m_h = 1 \text{ and } d_h \leq 15 \\ 1 & m_h = 1 \text{ and } d_h > 15 \\ m_h - 1 & m_h \geq 2 \text{ and } d_h \leq 15 \\ m_h & m_h \geq 2 \text{ and } d_h > 15 \end{cases} \quad (5)$$

where m_{end} indicates the last month of the 3-month interval, and m_h and d_h indicate the month and date of harvesting, respectively, obtained from the crop calendar data (Sacks et al., 2010). The above equation represents the case in which the number of days in a month is 30 or 31 (another threshold value for d_h , 14, is used for February). The assumption implied by the equation above is that, once harvesting occurs before the end of a month, climate conditions during the subsequent portion of that month do not affect yields. The beginning month of the reproductive growth period (m_{start}) is computed based on the last month of the 3-month interval:

$$m_{\text{start}} = \begin{cases} m_{\text{end}} - 2 + 12 & 1 \leq m_{\text{end}} \leq 2 \\ m_{\text{end}} - 2 & 3 \leq m_{\text{end}} \leq 12 \end{cases} \quad (6)$$

The same method of determining the reproductive growth period globally is found in earlier work (Iizumi et al., 2013, 2016).

We used the 3-month average data of the temperature and

precipitation hindcasts derived from the individual GCMs at 3- and 6-month lead times. The hindcast data are interpolated to have the same resolution as the reanalysis data (1.125°). Although the number of ensemble members differs across the GCMs, we use the ensemble means of the GCMs (Table 1). For the within-season predictions, for instance, the hindcast data for the average climate condition in July–August–September in the harvesting year which is predicted in July in the same year are used as inputs to the yield models to predict the yield variability in locations where the modeled reproductive growth period extends from July to September. The hindcast data for the average climate condition in July–August–September in the harvesting year which is predicted in April in the same year are used to derive the pre-season predictions. These hindcast data are bias-corrected using the cumulative distribution function-based downscaling method (CDFDM; Iizumi et al., 2010, 2011), following Iizumi et al. (2013). The RGP average temperature and precipitation anomaly data for 1984–2010 calculated from the reanalysis data are used as the reference in the bias correction. The bias correction is conducted by location, climatic variable and GCM.

2.3. Multi-model ensemble methods

Two methods are used to generate the MME hindcast data of the RGP average temperature and precipitation anomalies. One method is the average method, in which the average of the five GCMs is calculated using even weights. Taking temperature as an example for explanatory purposes, the average method provides the MME annual temperature anomaly time series (ΔT^a) as follows:

$$\Delta T_{t,g,s}^a = \frac{1}{n} \sum_{i=1}^n \Delta T_{t,g,s}^i, \quad (7)$$

where the subscript i indicates the i -th GCM; n indicates the number of GCMs ($=5$); and ΔT indicates the RGP average temperature anomaly without bias correction (°C). The average method-based MME data are then bias-corrected using the CDFDM method, as was done for the individual GCMs. The MME annual precipitation anomaly data are obtained in a similar manner.

Another method is the mosaic method. In this method, we select the best-performing single GCM for each location and cropping season (i.e., the reproductive growth period) based on the skill scores of the individual GCMs for yield variability and obtain the MME temperature anomalies (ΔT^m) as follows:

$$\Delta T_{t,g,s}^m = \Delta T_{t,g,s}^i \text{ when } \text{ROC}_i = \max\{\text{ROC}_j; j = 1, \dots, n\}, \quad (8)$$

where ROC_i indicates the relative operating characteristic (ROC) score (Wilks, 2006) of the i -th GCM that yields the highest score value among the five GCMs, and ΔT indicates the bias-corrected RGP average temperature anomaly (°C). Mathematically speaking, this selection is the same as assigning a weight of one to the best-performing single GCM and weights of zero to the remaining GCMs in the conventional MME technique using multiple linear regression (e.g., Krishnamurti et al., 2000).

2.4. Skill score

We use ROC scores (Wilks, 2006) to assess prediction skill. ROC scores equal to or greater than 0.6 indicates skill that is better than guessing, whereas those between 0.5 and 0.6 are associated with predictions that are little better than guesswork (Met Office, 2013). Our yield prediction presented here is deterministic even though multiple GCMs are used. Therefore, our predictions are evaluated as a binary classification tool. We test whether our predictions are true or false when they indicate yield gains (positive yield anomalies). Similarly, we assess whether our predictions are true or false when they indicated yield losses (negative yield anomalies). When a 2×2 contingency table

Table 2
A summary of the crop hindcast experiments.

Experiment	Climate data	Lead time (months)	Model type	Period	Reference	Notes
A	JRA-25 reanalysis	0	T-S, T-P, T and P	1984–2010	This study	The LTO cross-validation is used.
B	JRA-25 reanalysis	0	T-S	1983–2006	Iizumi et al. (2013)	Same as for A, but the leave-one-out method is used instead.
C	GRASP meteorological forcing dataset	0	T-P	1984–2010	This study	Same as for A.
D	Hindcasts from the individual GCMs (APCC, MSC-CANCM3, NCEP, NASA and PNU)	3 and 6	T-P	1984–2010	This study	The LTO cross-validation method is used for the yield models. The climate hindcast data are processed using the LTO bias-correction.
E	Hindcasts from a single GCM (SINTEX-F1)	3 and 6	T-S	1983–2006	Iizumi et al. (2013)	Same as for D, but the leave-one-out method is used instead.
F	MME hindcasts using the average method (MMEA) and the mosaic method (MMEB)	3 and 6	T-P	1984–2010 (grid level); 1984–2015 (national level)	This study	Same as for D.

is derived to compute ROC score for yield gain, the prediction for a year is categorized into “true positive” when both predicted and observed yield anomalies for that year are positive, whereas it is classified into “false positive” when a positive yield anomaly is predicted and a negative yield anomaly is observed (this corresponds to nonprobabilistic forecasts of discrete predictands – yield gain or not –). When computing ROC score for yield loss, another category – yield loss or not – is used to derive a contingency table. The ROC scores for the yield gains and yield losses are calculated separately and averaged for use as an indicator of the overall prediction skill for yield variability. The ROC scores are computed at a grid-cell level using the predicted grid-cell yield anomaly data. The national average yield anomaly time series calculated using the grid-cell prediction data are used when calculating ROC scores at the national level. For consistent comparison, the ROC scores are computed using the data presented in Iizumi et al. (2013) because a different skill score (the coefficient of determination) is used in Iizumi et al. (2013). The significance of the ROC scores was tested using 500 bootstrap replications, with a null hypothesis that the ROC score was equal to or lower than 0.5. The use of more categories (e.g., below-, near- and above-normal) is expected to be more informative. However, the limited sample size of the yield data prevents us from using more categories because at least 90 samples are generally required to properly estimate a 3×3 contingency table consisting of observation-forecast pairs (WMO, 2002).

2.5. Experimental design

Table 2 summarizes the crop hindcast experiments presented in this study. We first assess the prediction skill when the reanalysis data for the period 1984–2010 are used (A in Table 2), which theoretically represents the upper limit of prediction skill using climate model outputs (Challinor et al., 2005). For this assessment, we used the leave-three-out (LTO) cross-validation technique to avoid the possible influences of adjacent years in the temporarily correlated time series (Livezey, 1999) even though Iizumi et al. (2013) utilizes the leave-one-out cross validation technique. We remove three consecutive samples from the yield anomaly data and estimate the regression coefficients. The yield anomaly value in the middle of the year that is removed from the calibration subset is predicted, and this exercise is repeated for the sequential removal of all 27 samples. In this exercise, the single set of regression coefficients that give the highest likelihood is used to predict the value of the removed sample. The LTO cross-validation is completed for the four yield model types (the T-P, T-S, T and P models). We also use the data presented in Iizumi et al. (2013) as a reference; these data are based on reanalysis data and cover a relatively short period (1983–2006) and were generated using the T-S model (B in Table 2). We also examined whether the use of a reanalysis or climate observations influences the prediction skill using the Global Risk Assessment

toward Stable Production of Food (GRASP) project meteorological forcing dataset (GFD; Iizumi et al., 2014a), which is composed of bias-corrected JRA-25 reanalysis data, allowing the monthly mean temperature and precipitation of the GFD data to follow the Climate Research Unit (CRU) University of East Anglia data (Harris et al., 2014) (C in Table 2).

Second, we assess the prediction skill when the bias-corrected hindcast data of the individual GCMs for the period 1984–2010 are used (D in Table 2). The assessment is completed for the within-season and pre-season predictions. For this assessment, we use the LTO bias-correction technique and the LTO cross-validation technique. For each GCM, we remove three consecutive samples of the temperature and precipitation anomalies and derive the corresponding information on the GCM biases using the CDFDM method. The removed sample values for the temperature and precipitation anomalies are corrected based on this information and used to predict a yield anomaly value in the middle of the year from which the climate samples were removed. This exercise is repeated to complete all 27 samples. A set of the regression coefficients used earlier (A in Table 2) is used for this exercise (i.e., the yield sample in which the climate sample was removed was not used to estimate the regression coefficients). ROC scores are computed for the individual GCMs. This exercise is performed only for the T-P model, although we use the data produced using the T-S model and presented in Iizumi et al. (2013) as a reference; these data are based on the outputs of a single GCM (SINTEX-F1) (E in Table 2).

Third, we assess the prediction skill when the T-P model and each of the MME hindcast data generated from the average and mosaic methods are used (F in Table 2). In the mosaic method, the ROC scores of the individual GCMs calculated earlier are used to select the best-performing GCM. Note that the ROC scores used to perform the selection are based on the LTO bias-correction and the LTO cross-validation; therefore, our assessment is based on an independent validation subset.

Lastly, we calculate the national average yield anomalies for the period 1984–2015 using the within-season and pre-season grid-cell prediction data derived from the two MME datasets to evaluate the prediction skill at the national level (F in Table 2). The predicted national average yield anomalies are compared with the reported data from the Food and Agriculture of the United Nations (FAO), which has a longer temporal coverage than the gridded yield data.

3. Results

3.1. The upper limits of the prediction skill

Our results showed that the yield variability over 26–39% of the global harvest area was reliably explained by the RGP average temperature and soil moisture anomalies with crop variations (JRA25-T-S in Fig. 1). The extent of the area with a good prediction skill for yield

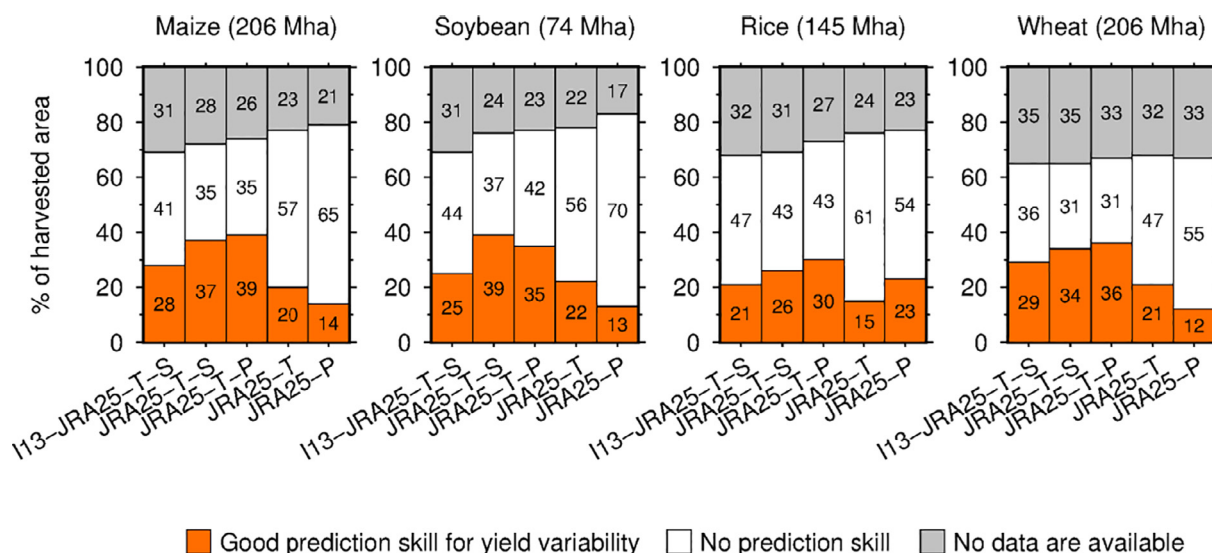


Fig. 1. The extent of areas where the observed yield variability in maize, rice, wheat and soybean is reliably explained by the reanalysis data (i.e., the ROC scores are significant at the 10% level). These quantities are expressed as percentages of the global harvested area in 2000 (Monfreda et al., 2008). The data are derived from the earlier assessment that covers the period of 1983–2006 (Iizumi et al., 2013) and uses the temperature-soil moisture model (I13-JRA25-T-S) and from the assessment that covers the period of 1984–2010 conducted in this study using the temperature-soil moisture model (JRA25-T-S), the temperature-precipitation model (JRA25-T-P), temperature-only model (JRA25-T) and the precipitation-only model (JRA25-P).

variability (i.e., ROC scores are significant at the 10% level) is hereafter referred to as an area share. The area shares are relatively higher than those calculated from the data presented in Iizumi et al. (2013) (21–29%; I13-JRA25-T-S in Fig. 1). The use of different yield dataset versions and study periods may partially explain these discrepancies. Importantly, a similar area share is obtained when the reanalysis precipitation data are used instead of the reanalysis soil moisture data (30–39%; JRA25-T-P in Fig. 1). The skill levels when using temperature and precipitation from the reanalysis data are comparable to those when using the climate observations (Fig. S1 in Supplementary materials). These results encourage us to use temperature and precipitation hindcast data for within-season and pre-season predictions.

As expected, the area shares decrease substantially when only the reanalysis temperature is used to explain the observed yield variability (15–22%; JRA25-T in Fig. 1). The area shares further decrease for all crops considered here (except rice) if only the reanalysis precipitation is used (12–23%; JRA25-P in Fig. 1). These results indicate that temperature, precipitation and the combination of these variables are important when explaining the observed yield variability in many parts of the world. Although the general expectation is that precipitation variability becomes less important with an increase in irrigation area, no clear relationship between the prediction skill and the irrigation area emerged in our results (Fig. S2).

3.2. The reliability of the within-season predictions

The areas where reliable within-season predictions can be made using the temperature and precipitation hindcast data from the individual GCMs are 7–18% for maize, 6–12% for soybean, 7–11% for rice and 9–18% for wheat; these quantities vary by GCM (APCC-T-P, MSC-CANCM3-T-P, NASA-T-P, NCEP-T-P and PNU-T-P in Fig. 2). The area shares are roughly comparable across the crops. No single best GCM is found. These area shares are comparable to those calculated using the data shown in Iizumi et al. (2013) (18%, 25%, 10% and 15% for maize, soybean, rice and wheat, respectively; I13-SINTEXF1-T-S in Fig. 2) except soybean. As the grid cells with extensive soybean harvested areas are concentrated in a few countries (the US, China, Brazil

and Argentina), small differences in the prediction skill in grid cells that locate within these countries may lead to large differences in the area share.

Surprisingly, the area shares are dramatically different between the two MME methods. The mosaic method showed that the yield variability over 25–38% of the global harvested area is reliably predictable three months before harvesting (MEM-T-P in Fig. 2). The corresponding value for the average method is 9–15% (MMEA-T-P in Fig. 2). The area shares of the mosaic method are 2.4–3.0 times larger than those of the average method and vary by crop. These results showed that the use of the average method-based MME hindcast data does not necessarily outperform the individual GCMs. For both MME methods, the crops and regions with good prediction skill include maize in Argentina and Indonesia, soybean in Argentina and South Africa, rice in Indonesia and Brazil and wheat in Australia and Northeast China (Figs. 3 and 4). The ROC scores of the mosaic method are almost always higher than those of the average method, regardless of the geographic region or crop considered. Again, there is no single best GCM when the mosaic method is utilized (Fig. S3).

Given the large area shares, the mosaic method allows us to derive reliable within-season predictions of yield variability at the national level in many crop-producing countries. Taking maize as an example for explanatory purposes, reliable within-season predictions of national yield variability can be produced in 42 countries or 36% of the 117 maize-producing countries using the mosaic method (Fig. 5). These countries include some major producers, such as Argentina, China, the US and France (Fig. 6). The corresponding values are 24% of the 29 soybean-producing countries, 25% of the 83 rice-producing countries and 30% of the 90 wheat-producing countries (Fig. 5). Soybean in Argentina, Paraguay and Italy, rice in Myanmar, and wheat in Australia, Argentina, Turkey, India and Russia are examples of pairs of crops and countries with good prediction skill at the national level (Fig. 6). The results derived using the mosaic method are better than those presented in Iizumi et al. (2013) (12–16%; I13-SINTEXF1-T-S in Fig. 5). However, caution is necessary because the magnitudes of the predicted national average yield anomalies are not comparable to those of the FAO-reported data (FAO, 2017), although the occurrence of yield gains or

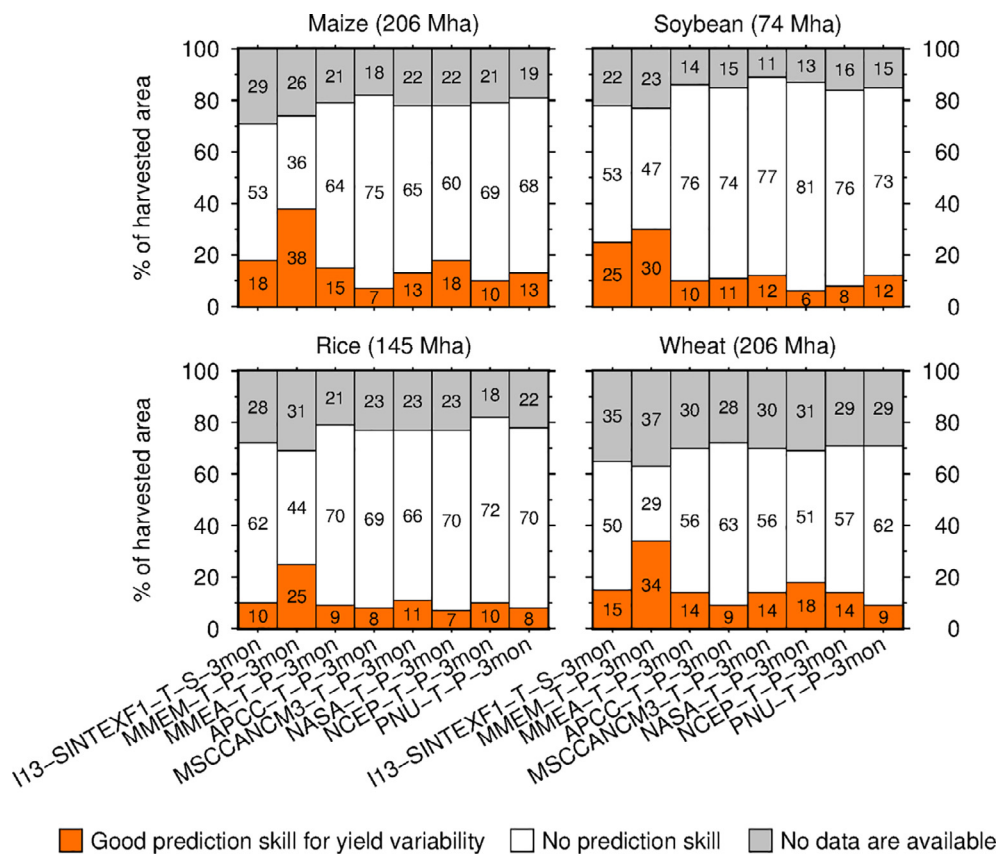


Fig. 2. The extent of areas where reliable within-season predictions of yield variability can be made using seasonal temperature and precipitation hindcast data (i.e., the ROC scores are significant at the 10% level). These quantities are expressed as percentages of the global harvested area in 2000. The data are derived from the earlier assessment that covers the period of 1983–2006 (Iizumi et al., 2013) and uses the temperature-soil moisture model (I13-SINTEXF1-T-S) and from the assessment that covers the period of 1984–2010 conducted in this study by applying the temperature-precipitation model to the five GCMs (APCC, MSCCANCNM3, NASA, NCEP and PNU) and the MME data derived using the two methods (the mosaic method, MMEM, and the average method, MMEA).

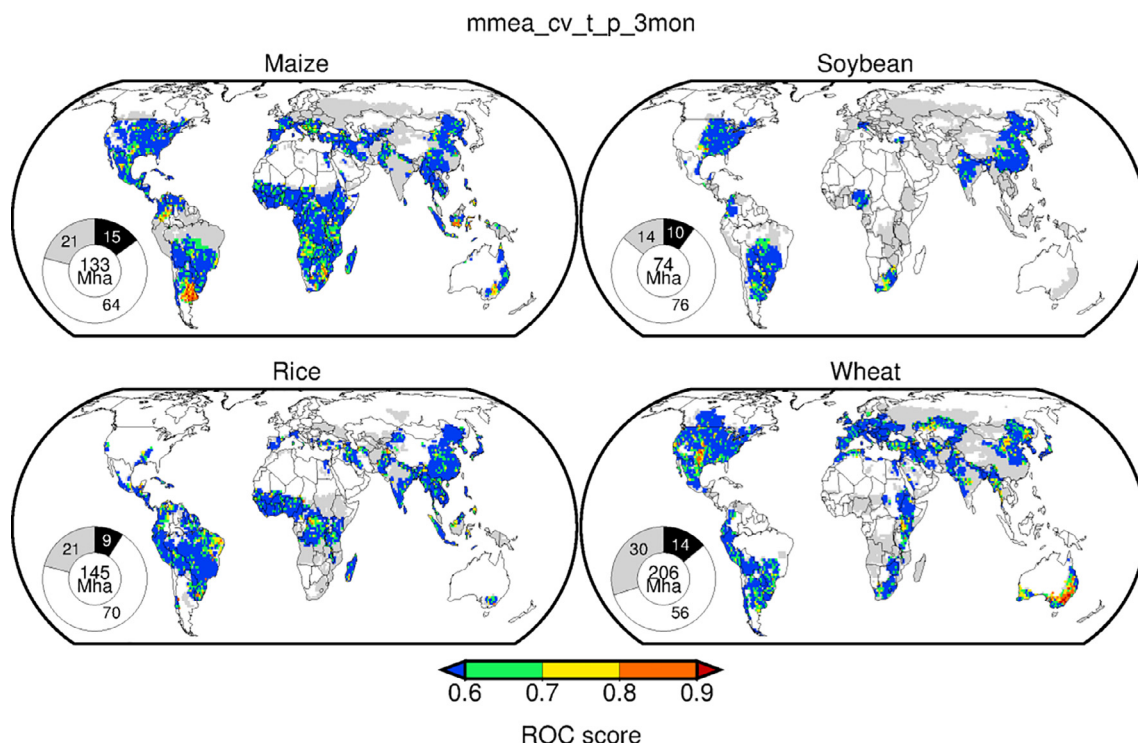


Fig. 3. The prediction skills of the within-season predictions of yield variability for maize, rice, wheat and soybean using the average method-based MME temperature and precipitation hindcast data. The pie diagrams indicate the percentage of the harvested area with a given skill score level (black – ROC score is significant at the 10% level, indicating that the within-season prediction is reliable; white – ROC score is nonsignificant, suggesting that the within-season prediction is less reliable; and gray – no hindcast is produced due to the lack of yield data). All of the data in the pie diagrams are normalized against the world production in 2000.

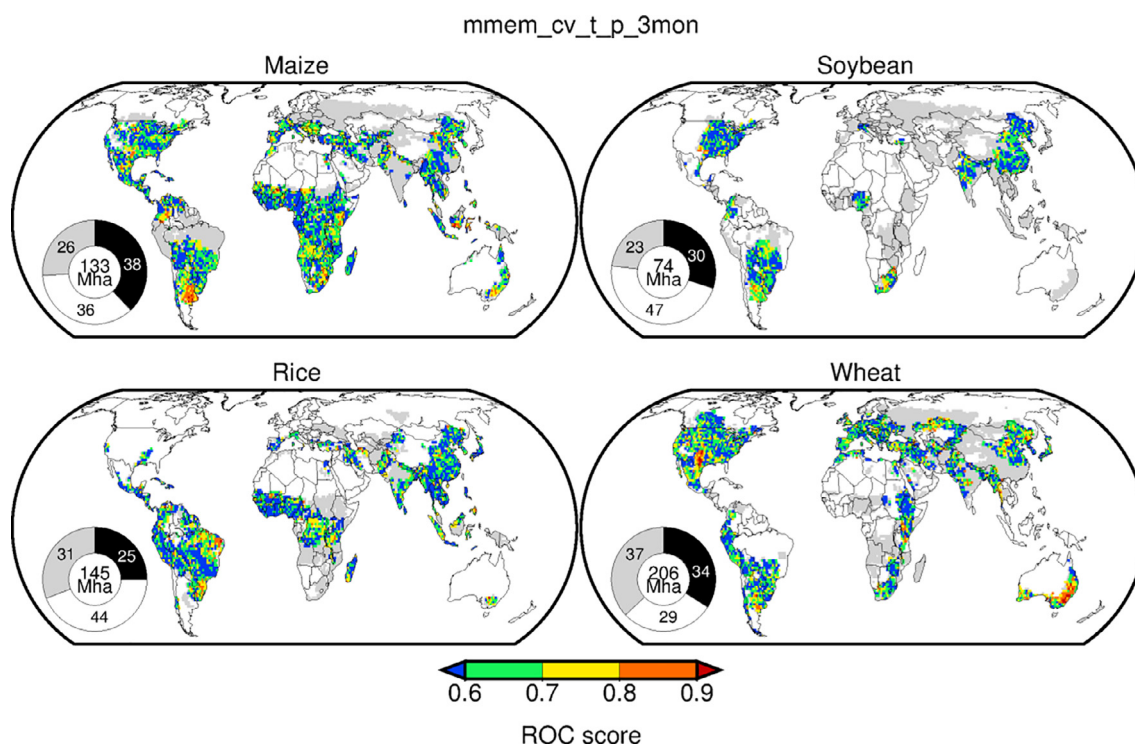


Fig. 4. Same as Fig. 3, but for the within-season predictions of yield variability using the mosaic method-based MME temperature and precipitation hindcast data.

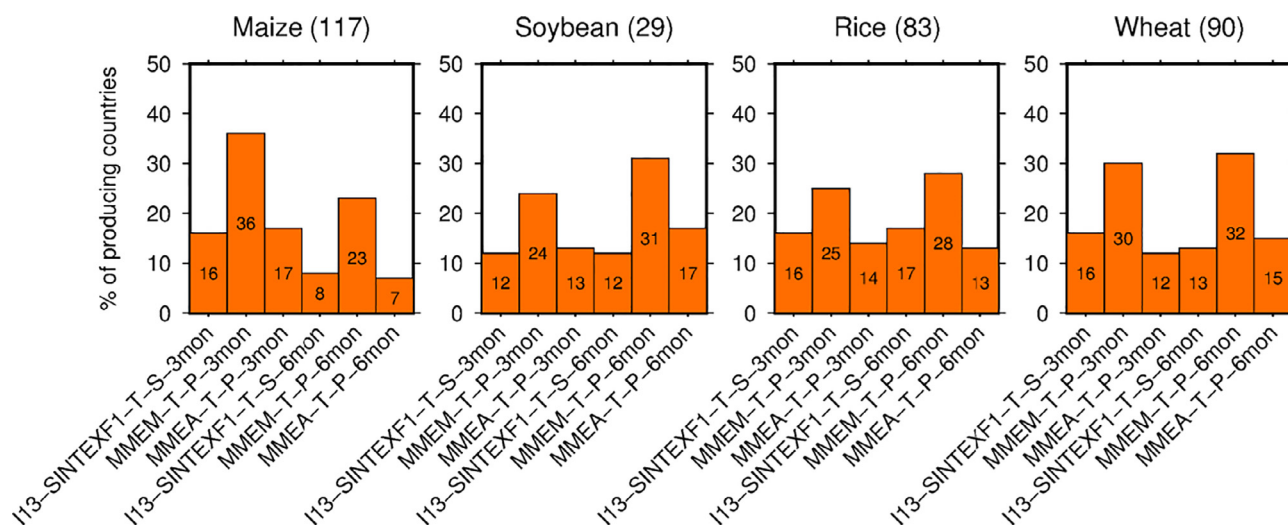


Fig. 5. Number of countries where reliable within-season and pre-season predictions of national yield variability are possible (i.e., the ROC scores are significant at the 10% level). The data are presented as percentages of the crop-producing countries. The numbers of crop-producing countries where the crops of interest are cultivated over 10 years during the study period are shown in parentheses.

yield losses is reliably predicted in many cases (Fig. 7). The countries where the average method reliably predicts the national yield variability are only 12–17% of the crop-producing countries (Fig. 5), indicating that the mosaic method outperforms the average method.

3.3. The reliability of the pre-season predictions

The area shares account for 28–41% of the global harvested area when the pre-season predictions based on the mosaic method are

considered (MMEM-T-P in Fig. S4). These values are almost comparable to those of the within-season predictions. However, this is mainly because we assess here is the sign of the yield change between two consecutive years, not the magnitude of the yield change. As previously shown for the within-season predictions, no single best GCM is found when the mosaic method is utilized for the pre-season predictions (Fig. S5).

The percentages of the crop-producing countries with good prediction skill are roughly comparable between the within-season and pre-

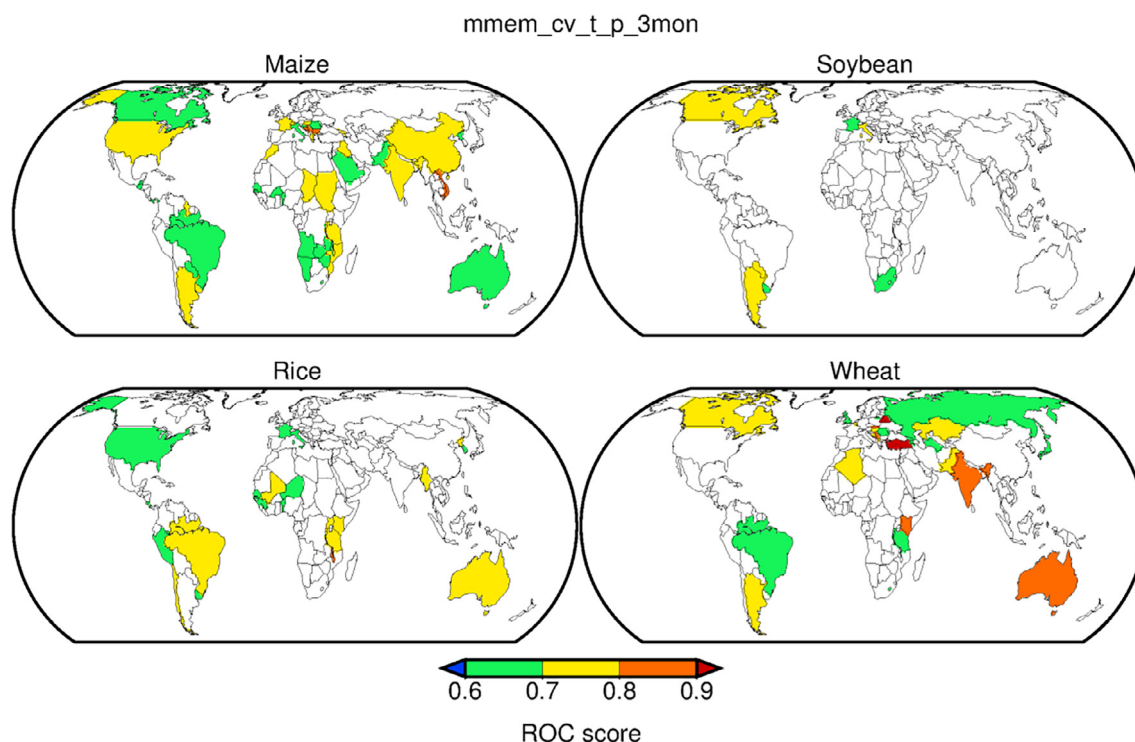


Fig. 6. The ROC scores of the within-season predictions of national yield variability for maize, rice, wheat and soybean using the mosaic method-based MME temperature and precipitation hindcast data. Only countries where the national ROC scores are significant are presented in color.

season predictions, with some variations by crop. In 23–32% of the crop-producing countries, the pre-season predictions made using the mosaic method reliably capture the yield variability six months before harvesting (MMEM-T-P in Figs. 5 and S6). In the pre-season predictions, the mosaic method also outperforms the results presented in Iizumi et al. (2013) (8–17%; I13-SINTEXF1-T-S in Fig. 5) and the average method (7–17%; MMEA-T-P in Fig. 5). This tendency is consistent between the within-season and pre-season predictions.

4. Discussion

The mosaic method, which uses a multi-model ensemble, represents important progress in global yield forecasting research beyond those of earlier works that were based on a single GCM (Iizumi et al., 2013). The results also show that the mosaic method outperforms the average method. Although the average method is found to have benefits when improving ROC scores for both temperature and precipitation compared to individual GCMs (Fig. S7), the benefits of the average method for yield predictions are smaller than those for the mosaic method. The mosaic method leads to increased area shares, which allows us to predict national yield variability in many more countries than before; this is a variable of interest for commodity and food security specialists.

After the food crises occurred in 2007/2008 and 2011, two major tools for monitoring global agriculture were launched and have been operated to aid food agencies worldwide in responding more efficiently to such shocks, based on the Group of Twenty (G20) Cannes Summit Final Declaration (G20, 2011). One is food price monitoring based on governmental agricultural statistics, as represented by the Agricultural Market Information System (AMIS) (Delincé, 2017). Another is crop condition monitoring using satellite remote sensing, as exemplified by the Global Crop Monitor (GEOGLAM, 2017). The contributions from satellite data and food price statistics in monitoring food insecurity are tremendous (Brown, 2016). However, as the main objective of these tools is monitoring, global yield forecasting based on seasonal climate forecasts represents an independent and complementary source of information.

Efforts to link predictable climate oscillations with early actions have been made by the FAO (2016b). The global forecast of yield impacts due to El Niño, with a lead time of 1–6 months, issued by the Ministry of Agriculture, Forestry and Fisheries of Japan in July 2014 (Iizumi, 2014d) is another example. The findings of this study offer a basis to develop an operational global yield forecasting service at climate centers and research institutes. Such a service is expected to have utility, as most yield forecasts that are currently in use are largely based on weather observation, satellite remote sensing and field surveys (Rembold et al., 2013; FAO, 2016a). Yield forecasts for maize and soybean, which are issued monthly (August through November) by the National Agricultural Statistics Service at the U.S. Department of Agriculture, are based on grower interviews and field surveys (USDA, 2012), and seasonal climate conditions for the period from the issue of yield forecast to harvesting are not considered. Monthly forecasts of the global supply and demand of major crops available from the World Agricultural Outlook Board are based on beginning stocks, production, domestic use, trade, and end-of-season carry-out stocks (USDA, 2018). Therefore, the effects of climate impacts on prospective yields (even the sign of the yield change relative to the previous year) can be useful. In addition, the reliability of seasonal climate forecast data (precipitation in particular) requires us to translate the data into yield impacts in a qualitative manner rather than in a quantitative manner.

Some possibilities for further improvements are as follows. First, many more GCMs than we used here are available at APCC, suggesting that the extent of areas with good prediction skill may increase further. Second, the yield responses to temperature and precipitation change with time due to technological improvements (Hawkins et al., 2013; Leng, 2017). At the same time, crops have experienced increasing exposure to weather extremes (Gourdji et al., 2013), and climate conditions could exceed those used in the model calibration. As changes in yield variability have been reported (Hawkins et al., 2013; Osborne and Wheeler, 2013; Iizumi and Ramankutty, 2016; Leng, 2017), statistical yield models must be recalibrated regularly to account for these changes. Third, various model structures are used in statistical yield modeling (e.g., Schauburger et al., 2017), although a single-model

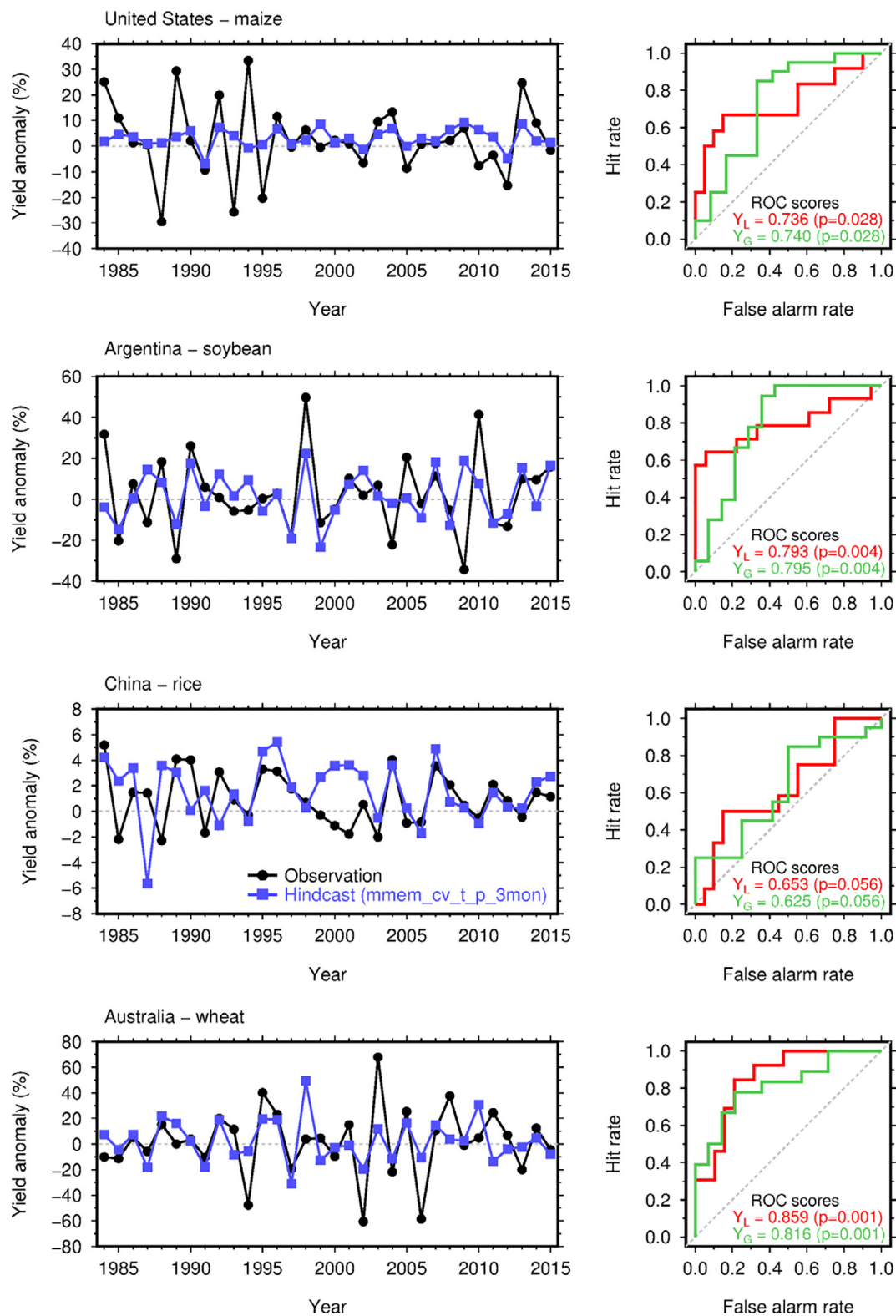


Fig. 7. Comparisons between the FAO-reported and within-season predictions of national average yield anomalies for the selected major producers from 1984 to 2015 (left panels) and the ROC curves calculated for yield gains (Y_G) and yield losses (Y_L) at various yield anomaly threshold settings (right panels). The ROC scores and their p-values are also shown. The results from the mosaic method-based MME hindcast data are presented.

structure is examined here because demonstrating the benefits of a multi-model ensemble is the main purpose of this study. Finally, not quantifying the uncertainty in the predicted yield variability is a major limitation of the mosaic method. A potential method of addressing this limitation is to develop yield models that are independent of those used

here to generate multi-method ensemble yield forecasts. Oceanic and atmospheric indices (Iizumi et al., 2014c, Yuan and Yamagata, 2015, Anderson et al., 2017; Ceglar et al., 2017, Heino et al., 2018) and heat and drought indices (Zampieri et al., 2017; Schauburger et al., 2017, Ceglar et al., 2018), which are potentially available from seasonal

climate forecast data and can be linked to yield variability, may be candidates.

5. Conclusion

We demonstrate an improvement in the methodology used to carry out global yield forecasting. Combined use of the mosaic method proposed here and MME seasonal temperature and precipitation forecast data allows reliable predictions of the sign of the yield change between two consecutive years (yield variability) over a large portion of the global harvested area three to six months before harvesting. The increased extent of the areas with good prediction skill enables reliable within-season and pre-season predictions of national yield variability in many countries. As seasonal temperature and precipitation forecast data are available from many climate centers and research institutes, a global yield forecasting service based on the mosaic method can be implemented for operational purposes. Crop yield forecast information delivered by such services will certainly increase the ability of food agencies and commercial entities to address the risks to food supplies posed by climate variability and change.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.cliser.2018.06.003>.

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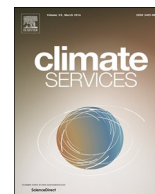
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